

Question [JEE Adv. 2016]

HW

[Ans. B,C,D]



Let $a, \lambda, \mu \in \mathbb{R}$. Consider the system of linear equations

$$ax + 2y = \lambda,$$

$$3x - 2y = \mu.$$

Which of the following statement(s) is(are) correct?

- A** If $a = -3$, then the system has infinitely many solutions for all values of λ and μ
- B** If $a \neq -3$, then the system has a unique solution for all values of λ and μ
- C** If $\lambda + \mu = 0$, then the system has infinitely many solutions for $a = -3$
- D** If $\lambda + \mu \neq 0$, then the system has no solution for $a = -3$

Question [JEE Mains 2025 (24 Jan. Shift 1)]

[Ans. A]



If the system of equations $2x - y + z = 4$, $5x + \lambda y + 3z = 12$, $100x - 47y + \mu z = 212$, has infinitely many solutions, then $\mu - 2\lambda$ is equal to:

A 57

B 59

C 55

D 56

Question [JEE Mains 2025]

[Ans. B]



If the system of linear equations $3x + y + \beta z = 3$; $2x + \alpha y - z = -3$; $x + 2y + z = 4$ has infinitely many solutions, then the value of $22\beta - 9\alpha$ is:

A 49

B 31

C 43

D 37

Question [JEE Mains 2023]

[Ans. C]



For $\alpha, \beta \in \mathbb{R}$, suppose the system of linear equations

$$x - y + z = 5$$

$$2x + 2y + \alpha z = 8$$

$$3x - y + 4z = \beta$$

has infinitely many solutions. Then α and β are the roots of:

A $x^2 - 10x + 16 = 0$

B $x^2 + 18x + 56 = 0$

C $x^2 - 18x + 56 = 0$

D $x^2 - 14x + 24 = 0$

Question [JEE Mains 2024 (Feb.-I)]

[Ans. B]



If the system of equations:

$$2x + 3y - z = 5$$

$$x + \alpha y + 3z = -4$$

$$3x - y + \beta z = 7$$

has infinitely many solutions, then $13\alpha\beta =$

A 1110

B 1120

C 1210

D 1220

Question [JEE Mains 2024 (Jan.-II)]

[Ans. 113]



Let for any three distinct consecutive terms a, b, c of an A.P., the lines $ax + by + c = 0$ be concurrent at the point P and $Q(\alpha, \beta)$ be a point such that the system of equations

$$x + y + z = 6$$

$$2x + 5y + \alpha z = \beta \text{ and}$$

$$x + 2y + 3z = 4,$$

SL

has infinitely many solutions. Then $(PQ)^2$ is equal to _____.